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Control Device

Abstract

A faceplate remote control device may be attached to a wall-mounted mechanical light switch that has a toggle actuator. The faceplate remote control device may include a toggle indicator that detects operation of the toggle actuator of the mechanical switch. The toggle indicator may cause the generation of an indication of detected operation of the toggle actuator. The toggle indicator may comprise a sliding member that is configured to move with the toggle actuator. The toggle indicator may comprise an obstruction detection device that includes an infrared (IR) transmitter and an IR receiver. The faceplate remote control device may include a control circuit and a wireless communication circuit. The control circuit may be configured to cause the wireless communication circuit to transmit one or more messages in response to detecting operation of the toggle actuator of the mechanical switch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/358,316 filed Jul. 25, 2023, which is a continuation of U.S. patent application Ser. No. 17/577,271 filed Jan. 17, 2022 (now U.S. Pat. No. 11,711,876, issued Jul. 25, 2023), which is a continuation of U.S. patent application Ser. No. 16/901,414 filed Jun. 15, 2020 (now U.S. Pat. No. 11,229,106, issued Jun. 15, 2020), which is a continuation of U.S. patent application Ser. No. 16/430,227, filed Jun. 3, 2019 (now U.S. Pat. No. 10,687,409, issued Jun. 16, 2020), which is a continuation of U.S. patent application Ser. No. 15/845,797, filed Dec. 18, 2017 (now U.S. Pat. No. 10,314,148, issued Jun. 4, 2019), which is a divisional of U.S. patent application Ser. No. 14/576,983, filed Dec. 19, 2014 (now U.S. Pat. No. 9,848,479, issued Dec. 19, 2017), which claims priority to U.S. Provisional Patent Application Ser. No. 61/920,865, filed Dec. 26, 2013, each of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Electrical loads, such as lamps, ceiling lighting fixtures, thermostats, shades, etc., may be controlled using load control devices. A load control device may be configured for wireless communication. For instance, a dimmer switch may be configured as a radio-frequency (RF) dimmer switch. Such a load control device may be associated with one or more devices in a load control system, such as a lighting control system. A load control device that participates in a load control system may receive wirelessly communicated messages (e.g., including commands) from one or more other devices of the load control system. The messages may cause the load control device to adjust the amount of power delivered to one or more electrical loads that are connected to the load control device.

[0003] FIG. 1 depicts an example prior art lighting control system **10** that includes a tabletop RF dimmer switch **20** and a lamp **30** that is plugged into the dimmer switch **20**, such that the dimmer switch **20** may be operated to control the amount of power delivered to the lamp **30**. The dimmer switch **20** may be electrically connected to an electrical circuit that includes an alternating-current (AC) power source **40** and an AC outlet **42** that is electrically connected to the AC power source **40**. The AC outlet **42** includes an upper switched receptacle **41** and a lower unswitched receptacle **43**. The electrical circuit further includes a wall-mounted light switch **46** that is mounted in an electrical wallbox and is coupled in series electrical connection between the AC power source **40** and the upper switched receptacle **41**. The lamp **30** may be controlled by the wall-mounted switch **46**. The dimmer switch **20** includes a plug **22** that is plugged into the switched receptacle **41**. The lamp **30** includes a plug **32** that is plugged into the plug **22** of the dimmer switch **20**, such that the delivery of AC power to the lamp **30** may be controlled by operating a toggle actuator (not shown)

of the wall-mounted switch **46** to open and close the switch.

[0004] The lighting control system **10** may further include one or more devices that are configured to wirelessly communicate with the dimmer switch **20**. As shown, the lighting control system **10** includes an occupancy and/or vacancy sensor **50**, a daylight sensor **60**, and a remote control device **70**, such as a remote keypad. One or more of the occupancy and/or vacancy sensor **50**, the daylight sensor **60**, and the remote control device **70** may wirelessly communicate with the dimmer switch **20** via RF signals **90**, for example to command the dimmer switch **20** to adjust the amount of AC power that is provided to the lamp **30**.

[0005] Control of the illustrated lighting control system **10** may be compromised when power is removed from the upper switched receptacle **41** of the outlet **42**. For instance, when the wall switch **46** is turned off, a wireless communication component of the dimmer switch **20**, such as a receiver, may be unpowered and thus unable to receive wirelessly communicated commands. This may undesirably render the dimmer switch **20** unresponsive to wirelessly communicated commands from the occupancy and/or vacancy sensor **50**, the daylight sensor **60**, and the remote control **70**, such as commands to turn on, turn off, or dim the lamp **30**.

[0006] Plugging the dimmer switch **20** into the lower unswitched receptacle **43** of the outlet **42** may ensure continuous power of the wireless communication component of the dimmer switch **20**, but would remove the ability to switch power to the lamp **30** using the wall-mounted switch **46**. This may be undesirable to a user of the lighting control system **10**. A user of the lighting control system **10** may prefer to be able to switch power to the lamp **30** via the wall-mounted switch **46**, while ensuring that the lamp **30** remains controllable by the dimmer switch **20**, for instance via one or more of the occupancy and/or vacancy sensor **50**, the daylight sensor **60**, and the remote control **70**.

[0007] Moreover, a user of the lighting control system **10** may be undesirably constrained from relocating the dimmer switch **20** and/or the lamp **30**. For example, if the user desires to move the dimmer switch **20** and the lamp **30** to a location wherefrom the electrical cord of the dimmer switch **20** will not reach the upper switched receptacle **41** of the outlet **42**, the user may be forced to plug the dimmer switch **20** into an unswitched outlet, such that the ability to switch the lamp **30** is lost, or may be forced to connect the dimmer switch **20** to the upper switched receptacle **41** using an extension cord, which may be impractical and/or aesthetically unpleasing.

SUMMARY

[0008] As described herein, a faceplate remote control device may be configured to be attached to a wall-mounted mechanical light switch that has a toggle actuator. The faceplate remote control device may include a housing that defines an opening that permits the toggle actuator of the light switch to protrude through the opening, such that the toggle actuator is operable when the faceplate remote control device is attached to the mechanical light switch.

[0009] The faceplate remote control device may be configured to detect operation of the toggle actuator of the mechanical switch, for example operation of the mechanical switch from a first position to a second position. The faceplate remote control device may include a toggle indicator that is configured to detect operation of the toggle actuator of the mechanical switch. The toggle indicator may cause the generation of an indication that operation of the toggle actuator is detected.

[0010] The toggle indicator may comprise a sliding member that is configured to move with the toggle actuator of the mechanical switch when the toggle actuator is operated. The sliding member may include an electrically-conductive wiper that is configured to abut a conductive pad when the toggle actuator is operated. Contact between the electrically-conductive wiper and the conductive pad may cause the conductive pad to generate an indication of operation of the toggle actuator.

[0011] The toggle indicator may comprise an obstruction detection device that includes an infrared (IR) transmitter and an IR receiver. The IR transmitter may generate an IR beam that is received at the IR receiver when the toggle actuator is in a first position. When the toggle actuator is operated, the IR beam may be obstructed, such that reception of the IR beam by the IR receiver is

interrupted. The IR receiver may generate a control signal that is representative of whether the IR beam is received, and thus representative of the position of the toggle actuator. The control signal may comprise an indication of when the toggle actuator is operated.

[0012] The faceplate remote control device may include a control circuit and a wireless communication circuit. The control circuit may be configured to cause the wireless communication circuit to transmit one or more messages in response to detecting operation of the toggle actuator of the mechanical switch. The one or more messages may be transmitted to one or more devices, such as a load control device, that are associated with the faceplate remote control device in a lighting control system. The one or more messages may include a command, such as a command that causes a load control device that is associated with the faceplate remote control device to adjust the intensity of a lighting load that is controlled by the load control device. The one or more messages may include, for example, a change of state signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 depicts a prior art lighting control system.

[0014] FIG. 2 depicts an example lighting control system that includes an example faceplate remote control device.

[0015] FIG. 3A is a perspective view of an example faceplate remote control device.

[0016] FIG. 3B is a cross-sectional view of the example faceplate remote control device depicted in FIG. 3A.

[0017] FIGS. 4A and 4B are perspective views of another example faceplate remote control device.

[0018] FIG. 5 is a simplified block diagram of an example faceplate remote control device.

[0019] FIG. 6 depicts another example lighting control system that includes another example faceplate remote control device.

DETAILED DESCRIPTION

[0020] FIG. 2 depicts an example load control system that is configured as a lighting control system **100**. The lighting control system **100** may include various components that are associated with each other, and that are configured to communicate with one another, for instance via wireless communication. The components of the lighting control system **100** may include, for example one or more load control devices, one or more electrical loads that are controlled via the one or more load control devices, one or more control devices (e.g., remote control devices) that are configured to control the load control devices, and/or one or more sensors that are configured to provide inputs (e.g., sensor readings) to the one or more load control devices.

[0021] As shown, the lighting control system **100** includes a controllable light source **110** and a faceplate remote control device **120** that may be configured to control the controllable light source **110**. The controllable light source **110** may include an integral lighting load (not shown) and an integral load regulation circuit (not shown). The controllable light source **110** and the faceplate remote control device **120** may include respective wireless communication circuits. For example, the controllable light source **110** may include a radio-frequency (RF) transmitter, and the faceplate remote control device **120** may include an RF transceiver. The faceplate remote control device **120** and the controllable light source **110** may be associated with one another, for example during a configuration procedure of the lighting control system **100**, such that the controllable light source **110** may be configured to respond to one or more messages transmitted by the faceplate remote control device **120**.

[0022] As shown, the controllable light source **110** may be installed in a table lamp **112**. The table lamp **112** may be plugged into a first electrical outlet **114** that has an upper switched receptacle **111** and a lower unswitched receptacle **113**. The lower unswitched receptacle **113** may be directly

coupled to an AC power source **102**, and the upper switched receptacle **111** may be coupled to the AC power source **102** through a standard wall-mounted mechanical switch **104** (e.g., a toggle switch or a standard light switch). The mechanical switch **104** may include a toggle actuator **106**. The mechanical switch **104** may be opened and closed in response to actuations of (e.g., operation of) the toggle actuator **106**. The mechanical switch **104** may comprise, for example, a maintained single-pole mechanical switch. The table lamp **112** may be plugged into the lower unswitched receptacle **113** of the electrical outlet **114**, such that the controllable light source **110** may be continuously powered from the AC power source **102**. The faceplate remote control device **120** may be operated to control the controllable light source **110**, without the need to plug the table lamp **112** into the upper switched receptacle **111** of the electrical outlet **114**.

[0023] The faceplate remote control device **120** may be configured to be attached to (e.g., mounted to) the mechanical switch **104**. For example, the faceplate remote control device **120** may be attached to the mechanical switch **104** in place of a standard faceplate or wall plate. In this regard, the faceplate remote control device **120** may replace a standard faceplate or wall plate that was previously attached to the mechanical switch **104**. The faceplate remote control device **120** may define an opening **122** through which the toggle actuator **106** of the mechanical switch **104** may protrude. As shown, the opening **122** may be configured to permit the toggle actuator **106** to protrude through the opening **122** such that the toggle actuator **106** is operable, for example by a user of the lighting control system **100**, and in particular a user of the faceplate remote control device **120**.

[0024] As shown, the controllable light source **110** includes a housing **115** (e.g., a glass housing) that defines a front surface **116**. The integral lighting load may be located within the housing **115** (e.g., surrounded by the housing **115**), and may be configured such that light generated by the integral lighting load shines out of the front surface **116** and/or the sides of the housing **115**. The front surface **116** of the housing **115** may be transparent or translucent, and may be dome shaped as shown, or flat. The integral lighting load of the controllable light source **110** may comprise, for example, an incandescent lamp, a halogen lamp, a compact fluorescent lamp, a light-emitting diode (LED) light engine, or other suitable light source.

[0025] The illustrated controllable light source **110** may also include an enclosure portion **118** to which the housing **115** may be attached, and a screw-in base (not shown) that may be attached to the enclosure portion **118**. The screw-in base may be configured to be screwed into a standard Edison socket, such that the controllable light source **110** is placed in electrical communication with (e.g., is electrically connected to) the AC power source **102**. Examples of screw-in luminaires are described in greater detail in commonly assigned U.S. Pat. No. 8,008,866, issued Aug. 30, 2011, entitled “Hybrid Light Source,” U.S. patent application publication no. 2012/0286689, published Nov. 15, 2012, entitled “Dimmable Screw-In Compact Fluorescent Lamp Having Integral Electronic Ballast Circuit,” and U.S. patent application Ser. No. 13/829,834, filed Mar. 14, 2013, entitled “Controllable Light Source,” the entire disclosures of which are incorporated herein by reference.

[0026] The integral load regulation circuit of the controllable light source **110** may be located within (e.g., housed inside) the enclosure portion **118**. The integral load regulation circuit may comprise, for example, a dimmer circuit, a ballast circuit, or an LED driver circuit, for controlling the intensity of the integral lighting load between a low-end intensity (e.g., approximately 1%) and a high-end intensity (e.g., approximately 100%). The controllable light source **110** may further include a control circuit (e.g., a microprocessor) and a wireless communication circuit (e.g., comprising an RF receiver) that may be housed inside the enclosure portion **118**. The control circuit may be configured to control the integral lighting load (e.g., via the integral load regulation circuit) in response to one or more messages that are received by the wireless communication circuit (e.g., via RF signals **108**), such as messages received from the faceplate remote control device **120**.

[0027] The faceplate remote control device **120** may be configured to operate as a state change

device. For example, the faceplate remote control device **120** may be configured to transmit one or more messages (e.g., digital messages) via wireless communication (e.g., via RF signals **108**) in response to actuations of the toggle actuator **106** of the mechanical switch **104**. The one or more messages may be indicative of a change of state within the lighting control system **100**. For example, one or more messages may be indicative of a change of state of the toggle actuator **106** of the mechanical switch **104**. Such messages may be referred to as change of state messages, or as change of state signals, and may be interpreted by one or more devices that are associated with the faceplate remote control device **120**, such as the controllable light source **110**, as indications (e.g., commands) to turn on, turn off, dim, etc. respective lighting loads. For example, the controllable light source **110** may cause the integral lighting load to turn on or off, or may cause the integral load regulation circuit to adjust an intensity of the integral lighting load, in response to the receipt of one or more messages transmitted by the faceplate remote control device **120** (e.g., via RF signals **108**). The one or more messages may be transmitted by the faceplate remote control device **120** in response to operation of the toggle actuator **106** of the mechanical switch **104**.

[0028] The lighting control system **100** may further include another load control device. For example, as shown, the lighting control system **100** further includes a plug-in load control device **130**. The plug-in load control device **130** is plugged into a second electrical outlet **136** that has two unswitched receptacles that are in electrical communication with the AC power source **102**. The lighting control system **100** further includes a floor lamp **132**. A standard light bulb **134** is installed in the floor lamp **132**. The floor lamp **132** is plugged into the plug-in load control device **130**, such that the plug-in load control device **130** may be operated to adjust the intensity of the light bulb **134** between a low end intensity (e.g., approximately 1%) and a high-end intensity (e.g., approximately 100%).

[0029] The plug-in load control device **130** may be associated with, and may be controlled by, the faceplate remote control device **120**. For example, the plug-in load control device **130** may cause the light bulb **134** to turn on or off, or may adjust an intensity of the light bulb **134**, in response to the receipt of one or more messages transmitted by the faceplate remote control device **120** (e.g., via RF signals **108**). This may allow the intensity of the light bulb **134** to be synchronized with that of the controllable light source **110**, for example. The plug-in load control device **130** may include one or more buttons (not shown) that are configured to provide local control of the plug-in load control device **130**, for example to allow adjustment of the intensity of the light bulb **134**. Alternatively, the plug-in load control device **130** may be a tabletop load control device or a wall-mounted dimmer switch.

[0030] The lighting control system **100** may further include a battery-powered handheld remote control device **140** that includes a plurality of buttons **142**. The handheld remote control device **140** may be configured to be mounted vertically to a wall, or to be supported on a pedestal that may be mounted on a tabletop. The handheld remote control device **140** may transmit one or more messages (e.g., via RF signals **108**) in response to operation of one or more of the buttons **142**. Examples of battery powered remote control devices are described in greater detail in commonly assigned U.S. Pat. No. 7,573,208, issued Aug. 22, 2009, entitled "Method Of Programming A Lighting Preset From A Radio-Frequency Remote Control," and U.S. Pat. No. 8,330,638, issued Dec. 11, 2012, entitled "Wireless Battery Powered Remote Control Having Multiple Mounting Means," the entire disclosures of which are incorporated herein by reference.

[0031] One or both of the controllable light source **110** and the plug-in load control device **130** may be configured to control the intensities of corresponding lighting loads (e.g., the integral lighting load and the light bulb **134**, respectively) in response to one or more messages received by the controllable light source **110** and the plug-in load control device **130**, for instance via RF signals **108**. Because the table lamp **112** is plugged into the lower unswitched receptacle **113** of the electrical outlet **114** as shown in FIG. 2, the controllable light source **110** may adjust the intensity of the integral lighting load independent of the position of the mechanical switch **104**. Accordingly,

the state of the controllable light source **110** (e.g., on or off) may be independent of a current position (e.g., closed or open) of the mechanical switch **104**.

[0032] The lighting control system **100** may further include one or more occupancy sensors **150** that are configured to detect occupancy and/or vacancy conditions in a space in which the lighting control system **100** is installed. Such an occupancy sensor **150** may transmit one or more messages (e.g., via RF signals **108**) to the controllable light source **110** and/or to the plug-in load control device **130**, in response to detecting the occupancy and/or vacancy conditions. Alternatively, the occupancy sensor **150** may operate as a vacancy sensor to turn off one or more lighting loads in response to detecting a vacancy condition (e.g., to not turn on the one or more lighting loads in response to detecting an occupancy condition). Examples of RF load control systems having occupancy and vacancy sensors are described in greater detail in commonly assigned U.S. Pat. No. 8,009,042, issued Aug. 30, 2011 Sep. 3, 2008, entitled "Radio Frequency Lighting Control System With Occupancy Sensing," U.S. Pat. No. 8,199,010, issued Jun. 12, 2012, entitled "Method And Apparatus For Configuring A Wireless Sensor," and U.S. Pat. No. 8,228,184, issued Jul. 24, 2012, entitled "Battery Powered Occupancy Sensor," the entire disclosures of which are incorporated herein by reference.

[0033] The lighting control system **100** may further include one or more remote daylight sensors **160** that are configured to measure a total light intensity in a space in which the lighting control system **100** is installed. Such a daylight sensor **160** may transmit one or more messages (e.g., via RF signals **108**) to the controllable light source **110** and/or to the plug-in load control device **130**. The one or more messages may include a measured light intensity, and may cause the controllable light source **110** and/or the plug-in load control device **130** to adjust the intensities of corresponding lighting loads (e.g., the integral lighting load and the light bulb **134**, respectively) in response to the measured light intensity. Examples of RF load control systems having daylight sensors are described in greater detail in commonly assigned U.S. Pat. No. 8,410,706, issued Apr. 2, 2013, entitled "Method Of Calibrating A Daylight Sensor," and U.S. Pat. No. 8,451,116, issued May 28, 2013, entitled "Wireless Battery-Powered Daylight Sensor," the entire disclosures of which are incorporated herein by reference.

[0034] In accordance with the illustrated lighting control system **100**, the faceplate remote control device **120**, the handheld remote control device **140**, the occupancy sensor **150**, and the daylight sensor **160** may operate as control-source devices (e.g., RF transmitters), and the controllable light source **110** and the plug-in load control device **130** may operate as control-target devices (e.g., RF receivers). It should be appreciated, however, that one or more of the control devices of the lighting control system **100** (e.g., all of the control devices) may comprise an RF transceiver, such that the control devices may be configured to both transmit and receive RF signals **108**. Examples of RF load control systems are described in commonly-assigned U.S. Pat. No. 5,905,442, issued on May 18, 1999, entitled "Method And Apparatus For Controlling And Determining The Status Of Electrical Devices From Remote Locations," and U.S. patent application Ser. No. 12/033,223, filed Feb. 19, 2008, entitled "Communication Protocol For A Radio Frequency Load Control System," the entire disclosures of which are incorporated herein by reference.

[0035] One of the load control devices (e.g., the controllable light source **110** or the plug-in load control device **130**) may be configured to operate as a control entity, such as a master device, within the lighting control system **100**. The master device may operate to at least partially control functionality of the other load control devices of the lighting control system **100**. The other load control devices of the lighting control system **100** may be configured to assume subservient roles to the master device (e.g., to operate as "slave" devices), such that the subservient devices will perform commands issued by the master device. It should be appreciated that if the lighting control system **100** includes only one load control device, the lighting control system **100** may not include a master device. One of the load control devices may be designated as the master device, for example by a user of the lighting control system **100**. Alternatively, one of the load control devices

may assume the role of the master device. For example, upon association with the lighting control system **100**, a load control device may poll the other load control devices of the load control system, for example via broadcast, to determine if the lighting control system **100** currently has a master device. If the polling load control device does not receive an answer that another device of the lighting control system **100** is the master device, the polling load control device may assume the role of the master device in the lighting control system **100**.

[0036] The master device may be configured to observe and/or record present state information pertaining to one or more subservient load control devices of the lighting control system **100**. In an illustrative example, with reference to the lighting control system **100** shown in FIG. 2, the plug-in load control device **130** may assume the role of the master device, and the controllable light source **110** may assume a subservient role to the plug-in load control device **130**, such that the plug-in load control device **130** is able to at least partially control operation of the controllable light source **110**. The plug-in load control device **130** may observe and/or record present state information, for example last known state information, pertaining to the controllable light source **110** (e.g., whether the integral lighting load of the controllable light source **110** is on or off).

[0037] The plug-in load control device **130**, in the role of the master device in the lighting control system **100**, may be configured such that if at least one lighting load, for example the light bulb **134** or the integral lighting load of the controllable light source **110**, is in an on state when the faceplate remote control device **120** transmits one or more change of state messages, the plug-in load control device **130** may cause one or more other lighting loads of the lighting control system **100** (e.g., each of the other lighting loads) to be operated from the on state to the off state, or left in the off state. For example, if the light bulb **134** is off and the integral lighting load of the controllable light source **110** is on when the toggle actuator **106** is operated, the faceplate remote control device **120** may transmit one or more change of state messages (e.g., via RF signals **108**) that may be received by the controllable light source **110** and the plug-in load control device **130**. The controllable light source **110** may ignore the one or more change of state messages from the faceplate remote control device **120**, for example in accordance with the subservient role the controllable light source **110** has with respect to the plug-in load control device **130**. When the one or more change of state messages are received by the plug-in load control device **130**, the plug-in load control device **130** may not change the state of the light bulb **134**, and may forward the one or more change of state messages to the controllable light source **110**. Alternatively, the plug-in load control device **130** may reference the last known state information pertaining to the controllable light source **110**, and may transmit an appropriate message (e.g., a command) to the controllable light source **110**, for example a command that causes the controllable light source **110** to turn the integral lighting load off. The controllable light source **110**, upon receipt of the forwarded change of message or the command, may change the state of the integral lighting load from on to off.

[0038] Alternatively, the plug-in load control device **130**, in the role of the master device in the lighting control system **100**, may be configured to maintain synchronization of the lighting loads of the lighting control system **100**. For example, if the state of the light bulb **134** is changed locally at the plug-in load control device **130**, the plug-in load control device **130** may transmit one or more messages (e.g., including a command) to the controllable light source **110** that cause the controllable light source **110** to change the state of the integral lighting load, thus keeping the states of the lighting loads of the lighting control system **100** synchronized to one another. If the state of the integral lighting load is changed locally at the controllable light source **110**, the plug-in load control device **130** may change the state of the light bulb **134**, thus keeping the states of the lighting loads of the lighting control system **100** synchronized to one another.

[0039] When the toggle actuator **106** of the mechanical switch **104** is actuated, the faceplate remote control device **120** may transmit one or more change of state messages that may be received by the controllable light source **110** and/or by the plug-in load control device **130**. The controllable light source **110** may ignore the one or more change of state messages from the faceplate remote control

device **120**, for example in accordance with the subservient role the controllable light source **110** has with respect to the plug-in load control device **130**. When the one or more change of state messages are received by the plug-in load control device **130**, the plug-in load control device **130** may change the state of the light bulb **134**, for example from on to off or from off to on, and may forward the one or more change of state messages to the controllable light source **110**.

Alternatively, the plug-in load control device **130** may transmit one or more messages (e.g., including a command) to the controllable light source **110**, for example to cause the controllable light source **110** to turn the integral lighting load on or off. The controllable light source **110**, upon receipt of the forwarded change of message or the one or more command messages, may change the state of the integral lighting load, for example from on to off or from off to on, such that synchronization of the lighting loads of the lighting control system **100** is maintained.

[0040] The role of the master device in a load control system in which the faceplate remote control device **120** is deployed, for instance the lighting control system **100**, need not be fulfilled by a load control device of the load control system, such as the controllable light source **110** or the plug-in load control device **130**. Such a load control system may include another device that is configured to fulfill the role of master device, for example a central controller, a main repeater, or the like. In such a configuration, one or more load control devices of the load control system, for example the controllable light source **110** and the plug-in load control device **130** of the lighting control system **100**, may be configured to assume subservient roles to the master device, and the master device may be configured to observe and/or record present state information pertaining to the subservient load control devices of the load control system. The subservient load control devices may be configured to ignore change of state messages transmitted by the faceplate remote control device **120**, and the master device may be configured to forward change of state messages received from the faceplate remote control device **120** to one or more of subservient load control devices, or may, upon receipt of one or more change of state messages from the faceplate remote control device **120**, transmit appropriate command messages to one or more of subservient load control devices.

[0041] A load control system in which the faceplate remote control device **120** is deployed, for instance the lighting control system **100**, need not include a central control entity, such as a master device. One or more load control devices that are associated with the lighting control system **100**, such as the controllable light source **110** and/or the plug-in load control device **130**, may be configured to be aware of present state information pertaining to one or more other load control devices of the lighting control system **100**. For example, the controllable light source **110** may be configured to be aware of whether the light bulb **134** controlled by the plug-in load control device **130** is on or off. Similarly, the plug-in load control device **130** may be configured to be aware of whether the integral lighting load of the controllable light source **110** is on or off. In such a configuration, the controllable light source **110** and the plug-in load control device **130** may operate to ensure that the lighting loads of the lighting control system **100** are kept synchronized with one another, for example responsive to one or more change of state messages that are transmitted by the faceplate remote control device **120**.

[0042] Alternative examples of state change devices are described in greater detail in commonly-assigned U.S. patent application Ser. No. 13/830,102, filed Mar. 14, 2013, entitled "State Change Devices For Switched Electrical Receptacles," the entire disclosure of which is incorporated herein by reference.

[0043] FIGS. 3A and 3B depict an example faceplate remote control device **200**. The faceplate remote control device **200** may be implemented, for example, as the faceplate remote control device **120** of the lighting control system **100** shown in FIG. 2. FIG. 3B is a simplified left side cross-sectional view of the example faceplate remote control device **200**, taken through the center of the faceplate remote control device **200**. The faceplate remote control device **200** includes a housing **211** that is configured to be attached to (e.g., mounted to) a wall-mounted mechanical switch, such as a standard wall-mounted light switch. As shown, the faceplate remote control

device **200** includes a two-part housing **211** that includes an adapter plate **212** that is configured to be mounted to a light switch (e.g., to a yoke of the light switch), and a front plate **210** that is adapted to be attached (e.g., semi-permanently attached) to the adapter plate **212**. In an example configuration, the front plate **210** and the adapter plate **212** may define respective attachment members that allow the front plate **210** to be secured to the adapter plate **212**. For example, the front plate **210** may define one or more resilient snap-fit connectors (not shown) that are designed to releasably engage within one or more complementary recesses (not shown) that are defined by the adapter plate **212**. It should be appreciated that the front plate **210** and/or the adapter plate **212** may be otherwise configured to be attachable to one another. It should further be appreciated that the faceplate remote control device **200** is not limited to the illustrated two-part housing **211**, and that the housing of the faceplate remote control device **200** may be alternatively configured, for instance as a one part housing. For example, the faceplate remote control device **200** may include a differently configured front plate (not shown) that defines one or more openings that are configured to receive fasteners (e.g., screws) so as to mount the faceplate remote control device **200** to a standard light switch.

[0044] As shown in FIGS. **3A** and **3B**, the faceplate remote control device **200** may be configured to be mounted to a standard wall-mounted mechanical switch **204** (e.g., the mechanical switch **104** of the lighting control system **100** shown in FIG. **2**). The faceplate remote control device **200** may be configured to maintain free operation of a toggle actuator **206** of the mechanical switch **204** when the faceplate remote control device **200** is attached to the mechanical switch **204**. For example, as shown, the housing **211** defines an opening **214**. In accordance with the illustrated faceplate remote control device **200**, the opening **214** is defined by the adapter plate **212** and the front plate **210**. The opening **214** may be configured to permit the toggle actuator **206** of the mechanical switch **204** to protrude through the opening **214** such that the toggle actuator **206** is operable, for example between respective first and second positions that correspond to open and closed positions of the mechanical switch **204**. The mechanical switch **204** may include a yoke **208** that allows the mechanical switch **204** to be mounted to a standard electrical wallbox, for instance using one or more mounting screws (not shown). The adapter plate **212** of the faceplate remote control device **200** may be configured to attach to the yoke **208** of the mechanical switch **204**, for example via one or more attachment screws (not shown).

[0045] The faceplate remote control device **200** may include a toggle indicator **220** that is configured to move along with the toggle actuator **206**. The toggle indicator **220** may be configured to generate one or more indications, for instance in response to operation of the toggle actuator **206**. As shown, the faceplate remote control device **200** includes a toggle indicator that is implemented as a sliding member **221**. The sliding member **221** may be configured to move along with the toggle actuator **206** when the toggle actuator **206** is operated.

[0046] As shown, the sliding member **221** may include a plate shaped body that defines a first end **219** that may be referred to as an upper end of the body, and an opposed second end **223** that is spaced from the first end **219**, and that may be referred to as a lower end of the body. The body of the sliding member **221** may define a length, for example as defined from the first end **219** to the second end **223**, such that the sliding member **221** at least partially covers the opening **214** as the toggle actuator is operated. For example, the sliding member **221** may be configured such that when the toggle actuator **206** is positioned in a first position that corresponds to a first limit of its travel (e.g., a fully up position as shown in FIGS. **3A** and **3B**), the second end **223** of the body is disposed below a lower edge of the opening **214**, and such that when the toggle actuator **206** is positioned in a second position that corresponds to a second limit of its travel (e.g., a fully down position), the first end **219** of the body is disposed above an upper edge of the opening **214**.

[0047] The sliding member **221** may be configured to captively attach to the toggle actuator **206** of the mechanical switch **204**, such that the sliding member **221** moves along with the toggle actuator **206** when the toggle actuator **206** is operated. For example, as shown, the sliding member may

define an aperture **222** that is configured to surround a portion of the toggle actuator **206**, such as an outer perimeter of the toggle actuator **206**. The housing **211** may define a cavity **213** within which the sliding member **221** may move relative to the housing **211** when the toggle actuator **206** is operated. In accordance with the illustrated faceplate remote control device **200**, the cavity **213** is defined by the front plate **210** of the housing **211**.

[0048] As shown, the sliding member **221** may be positioned adjacent to a rear surface of the front plate **210** when disposed in the cavity **213** (e.g., as shown in FIG. 3B). The sliding member **221** may be oriented in a plane that is parallel to an inner surface of the front plate **210**, and may be configured to move along a longitudinal direction relative to the faceplate remote control device **200** (e.g., in an up and down with respect to the orientation of the faceplate remote control device **200** as shown in FIGS. 3A and 3B). As shown, the sliding member **221** may be configured such that the aperture **222** surrounds a portion of the exterior of the toggle actuator **206**. The aperture **222** may be sized to be slightly larger than corresponding peripheral dimensions of the surrounded portion of the toggle actuator **206**, such that the sliding member **221** moves with the toggle actuator **206** (e.g., along the longitudinal direction) as the toggle actuator **206** is operated.

[0049] The faceplate remote control device **200** may also include a printed circuit board (PCB) **230** that may be housed inside the front plate **210**, for example disposed in the cavity **213**. Electrical circuitry of the faceplate remote control device **200** may be mounted to the printed circuit board **230**, and may include a control circuit, such as a microprocessor **232**. The sliding member **221** may include a wiper **224** that may be operable to contact (e.g., to abut) a front surface of the printed circuit board **230** when the toggle actuator is in a first position (e.g., as shown in FIG. 3B). The wiper **224** may be electrically conductive, and may contact one or more conductive pads and/or surfaces of the printed circuit board **230**, for example in accordance with a standard potentiometer configuration.

[0050] As the toggle actuator **206** of the mechanical switch **204** is operated, the sliding member **221** may move concurrently with the toggle actuator **206**, along the longitudinal direction, which may cause the wiper **224** to move across a corresponding front surface **231** of the printed circuit board **230**. For example, the front surface **231** of the printed circuit board **230** may include a conductive pad **234** that is communicatively coupled to (e.g., configured to transmit electrical signals to) the microprocessor **232**. When the toggle actuator **206** is in a first position (e.g., as shown in FIGS. 3A and 3B), the wiper **224** may be electrically coupled to the conductive pad **234**, and may generate a signal to the microprocessor **232** that the toggle actuator **206** is in the first position. In this regard, the sliding member **221** may be configured to cause the generation of a first indication when the toggle actuator **206** is operated into, and/or remains in, the first position.

[0051] When the toggle actuator **206** is operated to a second position, wherein the sliding member **221** moves along the longitudinal direction such that the wiper **224** is no longer electrically coupled to the conductive pad **234**, the microprocessor **232** may be configured to determine that the toggle actuator **206** is no longer in the first position. The second position may correspond, for example, to a down position of the toggle actuator **206**, or to an intermediate position between the up and down positions. In this regard, the sliding member **221** may be configured to cause the generation of a second indication when the toggle actuator **206** is operated out of the first position, or when the toggle actuator **206** is operated into the second position.

[0052] It should be appreciated that the printed circuit board **230** is not limited to the illustrated configuration having a single conductive pad **234**. For example, the front surface **231** of the printed circuit board **203** may alternatively include two or more conductive pads **234**, such that the wiper **224** may be electrically coupled to successive conductive pads **234** as the toggle actuator **206** is operated. In accordance with such a configuration, the microprocessor **232** may be configured to determine one or more intermediate or incremental positions of the toggle actuator **206** (e.g., between the up and down positions) as the toggle actuator **206** is operated. The faceplate remote control device **200** may alternatively include one or more mechanical tactile switches (not shown)

that may be mounted to the printed circuit board **230** (e.g., to the front surface **231**), and that may be actuated by the sliding member **221** when the toggle actuator **206** is in the up position or the down position.

[0053] The illustrated faceplate remote control device **200** may also include a wireless communication circuit **236**. The wireless communication circuit **236** may include, for example, an RF transmitter integrated circuit that is mounted to printed circuit board **230**, and an antenna (not shown). The antenna may comprise, for example, a loop antenna that is displaced on the printed circuit board **230**. The microprocessor **232** may be configured to cause the wireless communication circuit **236** to transmit one or more messages (e.g., state change messages), for instance in response to operation of the toggle actuator **206**. For example, a load control device that is associated with the faceplate remote control device **200** may be operable to control a corresponding electrical load (e.g., a lighting load) in response to one or more messages transmitted by the faceplate remote control device **200**.

[0054] The microprocessor **232** may be configured to detect one or more predetermined patterns of operation of the toggle actuator **206**. For example, the microprocessor **232** may be configured to detect one or more predetermined patterns of operation that comprise sequences of toggles of the toggle actuator **206**. A sequence of toggles of the toggle actuator **206** may comprise, for example, operating the toggle actuator **206** a predetermined number of times between the first and second positions, between the first position and an intermediate position, between the second position and an intermediate position, between respective first and second intermediate positions, or the like. A sequence of toggles of the toggle actuator **206** may include one or more temporal components. For example, an amount of time during which the toggle actuator **206** is left in particular position (e.g., the first and/or second positions) may distinguish a first sequence of toggles from a second sequence of toggles. One or more predetermined patterns of operation of the toggle actuator **206** may be configured, for example, by a user of the faceplate remote control device **200**.

[0055] Such predetermined patterns of operation of the toggle actuator **206** may be associated with desired functionality of one or more devices that are associated with the faceplate remote control device **200** in a lighting control system. For example, one or more predetermined patterns of operation of the toggle actuator **206** may be associated with the selection of corresponding lighting presets (e.g., lighting scenes) by a user of the faceplate remote control device **200**. To illustrate, a first predetermined sequence of toggles of the toggle actuator **206** may be used to select a first preset, a second predetermined sequence of toggles of the toggle actuator **206** may be used to select a second preset, and so on. In another example, one or more predetermined patterns of operation of the toggle actuator **206** may be associated with the selection of a fade rate of a lighting load by the user of the faceplate remote control device **200**. To illustrate, a first predetermined sequence of toggles of the toggle actuator **206** may be used to cause a lighting load to quickly turn on to full intensity, and a second predetermined sequence of toggles of the toggle actuator **206** may cause the lighting load to slowly fade to a lowest intensity (e.g., to off). Upon detecting a predetermined pattern of operation of the toggle actuator **206** that is associated with a selected lighting preset or a selected fade rate, the microprocessor **232** may cause the wireless communication circuit **236** to transmit one or more messages to one or more devices that are associated with the faceplate remote control device **200**. The one or messages may include, for example, commands that cause one or more load control devices that are associated with the faceplate remote control device **200** to adjust the intensities of corresponding lighting loads in accordance with the selected lighting preset or fade rate.

[0056] As shown, the faceplate remote control device **200** further includes a programming button **240**, which is mechanically coupled to a tactile switch **242** that is mounted to a rear surface **233** of the printed circuit board **230**. One or more devices, such as a load control device, may be associated with the faceplate remote control device **200**, for example in response to actuations of a button on the load control device and the programming button **240** of the faceplate remote control

device **200**. In this regard, the programming button **240** may be operated to initiate a process to associate the faceplate remote control device **200** with one or more devices, for instance one or more devices of a load control system, such as a lighting control system.

[0057] The faceplate remote control device **200** may further include a power source. The power source may include, for example, an energy storage device such as a coin cell battery **244**. The faceplate remote control device **200** may further include a battery holder **246** that is configured to secure the battery **244** in position relative to the faceplate remote control device **200**. As shown, the faceplate remote control device **200** may include a battery holder **246** that is located in the front plate **210**. When disposed in the battery holder **246**, the battery **244** may be electrically coupled to the printed circuit board **230**, and may provide power to the microprocessor **232** and/or to the wireless communications circuit **236**.

[0058] The faceplate remote control device **200** may include one or more other power sources, for instance in addition to, or in lieu of, the battery **244**. For example, the faceplate remote control device **200** may include a solar cell or photovoltaic coating, such as a photovoltaic film, (not shown) that may be displaced on (e.g., attached to) one or more surfaces (e.g., exterior surfaces) of the housing **211** of the faceplate remote control device **200**, such as on a front surface of the front plate **210**. The photovoltaic coating may be configured, for example, to charge the battery **244** or another energy storage device, such as a capacitor, and/or to directly power the microprocessor **232** and/or the wireless communication circuit **236**. In another example, the faceplate remote control device **200** may include a kinetic power source (not shown) that is configured to power the microprocessor **232** and/or the wireless communication circuit **236**. The kinetic power source may be configured to derive power from, for example, movements of the toggle actuator **206**.

[0059] The faceplate remote control device **200** may define a user interface. The user interface may be configured to receive one or more inputs from a user of the faceplate remote control device **200**. Such inputs may, for example, cause the faceplate remote control device **200** to issue commands to one or more devices that are associated with the faceplate remote control device **200**. As shown in FIG. 3A, the faceplate remote control device **200** defines a user interface that may include one or more buttons, such as buttons **250**, **252**. The buttons **250**, **252** may be configured to, upon actuation, cause the wireless communication circuit **236** to transmit one or more messages. The one or more messages may include, for example, one or more commands directed to one or more devices (e.g., load control devices such as lighting control devices) that are associated with the faceplate remote control device **200**. Such messages may be referred to as command messages.

[0060] The one or more command messages may provide advanced control of a lighting load that is controlled by the load control device. To illustrate, in a lighting control system in which the faceplate remote control device **200** is associated with the controllable light source **110**, one or more command messages transmitted in response to operation of one or more of the buttons **250**, **252** may cause the integral load regulation circuit of the controllable light source **110** to raise and/or lower the intensity of the integral lighting load. In accordance with an alternative configuration of the faceplate remote control device **200**, the buttons **250**, **252** may be configured to cause the selection of associated lighting scenes or lighting presets of the lighting control system **100**. The buttons **250**, **252** may be mechanically coupled to corresponding mechanical tactile switches (not shown) that may be mounted to the printed circuit board **230**. It should be appreciated that the user interface of the faceplate remote control device **200** is not limited to the illustrated mechanical buttons **250**, **252**. In an alternative configuration, the faceplate remote control device **200** may include a capacitive or resistive touch display (not shown), and the user interface may include one or more graphical representations of controls (e.g., soft buttons) exhibited (e.g., displayed) on the touch display.

[0061] As shown, the faceplate remote control device **200** may further include a visual display, such as a linear array of visual indicators **260** that may be illuminated to provide feedback, for instance feedback related to an intensity of a lighting load that is controlled by a load control that is

associated with the faceplate remote control device **200**. The visual indicators **260** of the linear array may be illuminated, for example, by light-emitting diodes (not shown) that are mounted on the printed circuit board **230**. Circuits for efficiently illuminating one or more light-emitting diodes are described in greater detail in commonly-assigned U.S. patent application publication no. 2012/0286940, published Nov. 12, 2012, entitled "Control Device Having A Nightlight," the entire disclosure of which is incorporated herein by reference. In an alternative configuration, the visual indicators of the faceplate remote control device **200** may be displayed on a capacitive or resistive touch display (not shown), such as a display that exhibits one or more soft buttons of a user interface of the faceplate remote control device **200**.

[0062] The faceplate remote control device **200** may further include a sensing device **270** that is configured to provide automated control of a lighting load that is controlled by a load control device that is associated with the faceplate remote control device **200**. The sensing device **270** may be mounted to the printed circuit board **230**, for example, and the microprocessor **232** may be configured to cause the wireless communication circuit **236** to transmit one or more messages to the load control device in response to the sensing device **270**. For example, the sensing device **270** may be an occupancy or vacancy sensing device, such that the microprocessor **232** may be configured to cause one or more messages to be transmitted to the load control device in response to the sensing device **270** detecting an occupancy or vacancy condition in a space around the faceplate remote control device **200**. The sensing device **270** may comprise, for example, a passive infrared (PIR) detector that is operable to receive infrared energy through a lens **272** located in the front plate **210** of the housing **211** of the faceplate remote control device **200**. Alternatively, the sensing device **270** may comprise an ultrasonic detector, a microwave detector, or any combination of passive infrared, ultrasonic, and/or microwave detectors. The load control device may turn the controlled lighting load on and off in response to the sensing device **270** of the faceplate remote control device **200** detecting occupancy and/or vacancy conditions, for example in a similar manner as the controllable light source **110** and/or the plug-in load control device **130** operate in response to messages received from the occupancy sensor **150**, as described herein.

[0063] The sensing device **270** may alternatively comprise a daylight sensing device that is configured to measure a light level in a space surrounding the faceplate remote control device **200**. The microprocessor **232** may be configured to cause the wireless communication circuit **236** to transmit one or more messages, for example including one or more light level measurements, to a load control device that is associated with the faceplate remote control device **200**. The one or more messages may be received by the load control device, and the load control device, responsive to receipt of the one or more messages, may adjust an intensity of a corresponding lighting load that is controlled by the load control device. The sensing device **270** may alternatively comprise a temperature sensing device that is configured to measure a temperature of a space surrounding the faceplate remote control device **200**. It should be appreciated that the faceplate remote control device **200** may include other types of sensing devices, or any combination of occupancy or vacancy sensing devices, daylight sensing devices, and/or temperature sensing devices.

[0064] It should be appreciated that while the faceplate remote control device **200** is illustrated in accordance with a single-gang configuration in FIGS. 3A and 3B, that the faceplate remote control device **200** may alternatively be configured in accordance with a multi-gang faceplate structure. For example, the faceplate remote control device may alternatively be configured to include two openings for receiving respective toggle actuators of two mechanical switches. In accordance with such a configuration, the faceplate remote control device may be configured to determine the respective positions of each toggle actuator, and to transmit one or more wireless messages in response to the positions of one or both of the toggle actuators. Alternatively, such a configuration of the faceplate remote control device may be configured to determine the position of a single one of the toggle actuators.

[0065] FIGS. 4A and 4B depict another example faceplate remote control device **300**. The faceplate

remote control device **300** may be implemented, for example, as the faceplate remote control device **120** of the lighting control system **100** shown in FIG. 2. The faceplate remote control device **300** includes a housing **311** that is configured to be attached to (e.g., mounted to) a wall-mounted mechanical switch, such as a standard wall-mounted light switch. As shown, the faceplate remote control device **300** includes a two-part housing **311** that includes an adapter plate **312** that is configured to be mounted to a light switch (e.g., to a yoke of the light switch), and a front plate **310** that is adapted to be attached (e.g., semi-permanently attached) to the adapter plate **312**. In an example configuration, the front plate **310** and the adapter plate **312** may define respective attachment members that allow the front plate **310** to be secured to the adapter plate **312**. For example, the front plate **310** may define one or more resilient snap-fit connectors (not shown) that are designed to releasably engage within one or more complementary recesses (not shown) that are defined by the adapter plate **312**. It should be appreciated that the front plate **310** and/or the adapter plate **312** may be otherwise configured to be attachable to one another.

[0066] The faceplate remote control device **300** may be configured to be mounted to a standard wall-mounted mechanical switch (e.g., the mechanical switch **104** of the lighting control system **100** shown in FIG. 2). The faceplate remote control device **300** may be configured to maintain free operation of a toggle actuator **306** of the mechanical switch when the faceplate remote control device **300** is attached to the mechanical switch. For example, as shown, the housing **311** defines an opening **314**. The opening **314** may be configured to permit the toggle actuator **306** of the mechanical switch to protrude through the opening **314** such that the toggle actuator **306** is operable, for example between respective first and second positions that correspond to open and closed positions of the mechanical switch. The mechanical switch may include a yoke that allows the mechanical switch to be mounted to a standard electrical wallbox, for instance using one or more mounting screws (not shown). The adapter plate **312** of the faceplate remote control device **300** may be configured to attach to the yoke of the mechanical switch, for example via one or more attachment screws (not shown).

[0067] The faceplate remote control device **300** may include a toggle indicator that is configured to generate one or more indications, for instance in response to operation of the toggle actuator **306**. As shown, the faceplate remote control device **300** includes a toggle indicator that is implemented as an obstruction detection device comprising an infrared (IR) transmitter **320** and an IR receiver **322**. The IR transmitter **320** and the IR receiver **322** may be disposed on opposed sides of the opening **314**, for instance mounted inside the front plate **310** on opposite sides of the opening **314**. As shown, the IR transmitter **320** may be disposed near a first side of the opening **314**, and the IR receiver **322** may be disposed near a second side of the opening **314** that is opposite the first side of the opening **314**.

[0068] The IR transmitter **320** may be configured to emit an IR beam **324**, and the IR receiver **322** may be configured to receive the IR beam **324**, when the toggle actuator **306** is in a first position, such as an up position as shown in FIG. 4A. When the toggle actuator **306** is operated into a second position, such as a down position of the toggle actuator **306**, the IR beam **324** may be obstructed, such that reception of the IR beam **324** by the IR receiver **322** is interrupted. The IR receiver **322** may generate a control signal that is representative of whether the IR receiver **322** receives the IR beam **324**. The control signal may thus be representative of the position of the toggle actuator **306**, and may comprise an indication of when the toggle actuator **306** is operated. The faceplate remote control device **300** may further include a control circuit (not shown) that is configured to determine the position of the toggle actuator **306** in response to the control signal generated by the IR receiver **322**. The faceplate remote control device **300** may further include a wireless communication circuit (not shown). The control circuit may be configured to cause the wireless communication circuit to transmit one or more messages in response to the position of the toggle actuator **306**. It should be appreciated that the faceplate remote control device **300** is not limited to the illustrated obstruction detection device. For example, the obstruction detection device may alternatively comprise

magnetic, fiber optic, or other proximity detection techniques to detect the position of the toggle actuator **306**.

[0069] FIG. 5 is a simplified block diagram of an example faceplate remote control device **400**. The faceplate remote control device **400** may be implemented, for example, as the faceplate remote control device **120** of the lighting control system **100** shown in FIG. 2, as the faceplate remote control device **200** shown in FIGS. 3A and 3B, and/or as the faceplate remote control device **300** shown in FIGS. 4A and 4B. The faceplate remote control device **400** may include a control circuit **410**. The control circuit **410** may include one or more of a processor (e.g., a microprocessor), a microcontroller, a programmable logic device (PLD), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any suitable processing device. The faceplate remote control device **400** may be mounted to a standard wall-mounted mechanical switch. The control circuit **410** may be configured to detect the position of a toggle actuator of the mechanical switch.

[0070] The faceplate remote control device **400** may comprise a toggle actuator detector circuit **412** that is communicatively coupled to the control circuit **410** and that is configured to generate a toggle actuator control signal VTOG that is representative of the position of the toggle actuator of the mechanical switch. The toggle actuator control signal VTOG may be received by the control circuit **410**, and may comprise an indication of when the toggle actuator of the mechanical switch is operated. The toggle actuator detector circuit **412** may be implemented as, for example, the wiper **224** of the sliding member **221** and the conductive pad **234** on the printed circuit board **230** of the faceplate remote control device **200**, as a mechanical tactile switch (not shown) that could be mounted to the printed circuit board **230** and that could be actuated by the sliding member **221**, and/or as the IR transmitter **320** and the IR receiver **322** of the faceplate remote control device **300**. The control circuit **410** may be configured to determine the position of the toggle actuator of the mechanical switch in response to the toggle actuator control signal VTOG.

[0071] The faceplate remote control device **400** may further include a wireless communication circuit **414** that is communicatively coupled to the control circuit **410**. The wireless communication circuit **414** may include, for example, an RF transmitter that is coupled to an antenna for transmitting RF signals. The control circuit **410** may be configured to cause the wireless communication circuit **414** to transmit one or more messages (e.g., via RF signals) in response to the position of the toggle actuator of the mechanical switch determined from the toggle actuator control signal VTOG. Alternatively, the wireless communication circuit **414** may include an RF receiver for receiving RF signals, an RF transceiver for transmitting and receiving RF signals, or an infrared (IR) transmitter for transmitting IR signals. For example, the control circuit **410** may be configured to receive one or more messages, via the wireless communication circuit **414**, which may include for example, an amount of power being delivered to an electrical load that is controlled by a load control device that is associated with the faceplate remote control device **400**.

[0072] The control circuit **410** may be configured to detect one or more predetermined patterns of operation of the toggle actuator of the mechanical switch, for instance as described herein. For example, the control circuit **410** may be configured to detect a predetermined pattern of operation that comprises one or more toggles of the toggle actuator between the first and second positions a predetermined number of times, for instance within a predetermined interval of time. Such predetermined patterns of operation of the toggle actuator may be associated with desired functionality of one or more devices that are associated with the faceplate remote control device **400** in a lighting control system. For example, one or more predetermined patterns of operation of the toggle actuator may be associated with the selection of corresponding lighting presets (e.g., lighting scenes) by a user of the faceplate remote control device **400**. Upon detecting a predetermined pattern of operation of the toggle actuator that is associated with a selected lighting preset, the control circuit **410** may cause the wireless communication circuit **414** to transmit one or more messages to one or more devices that are associated with the faceplate remote control device

400. The one or messages may include, for example, commands that cause one or more load control devices that are associated with the faceplate remote control device **400** to adjust the intensities of corresponding lighting loads in accordance with the selected lighting preset. [0073] The faceplate remote control device **400** may further include a memory **416**. The memory **416** may be communicatively coupled to the control circuit **410**, and may operate to store information, such as one or more lighting presets that may be associated with predetermined patterns of operation of the toggle actuator of the mechanical switch. The control circuit **410** may be configured to store such information in, and/or to retrieve such information from, the memory **416**. The memory **416** may include any component suitable for storing such information. For example, the memory **416** may include one or more components of volatile and/or non-volatile memory, in any combination. The memory **416** may be internal and/or external with respect to the control circuit **410**. For example, the memory **416** may be implemented as an external integrated circuit (IC), or as an internal circuit of the control circuit **410** (e.g., integrated within a microchip). [0074] The faceplate remote control device **400** may further include one or more buttons, such as one or more control buttons **418** and/or a programming button **420** that are communicatively coupled to the control circuit **410**, for instance such that the control circuit **410** may receive respective inputs from the one or more control buttons **418** and the programming button **420**. The faceplate remote control device **400** may further include a visual display **422** that is configured to provide feedback, for example of the amount of power being delivered to the electrical load being controlled by the load control device that is associated with the faceplate remote control device **400**. The visual display **422** may comprise, for example, one or more light emitting diodes (LEDs) illuminating a linear array of visual indicators on the faceplate remote control device **400**. The faceplate remote control device **400** may further include a sensing circuit **424** that comprises a sensing device. The sensing circuit **424** may be configured to provide automated control of the lighting load that is controlled by the load control device that is associated with the faceplate remote control device **400**. For example, the sensing circuit **424** may comprise an occupancy sensing circuit or a daylight sensing circuit (e.g., similar to the sensing device **270** of the faceplate remote control device **200** shown in FIGS. 3A and 3B). The faceplate remote control device **400** may further include an energy storage device, such as a battery **426** (e.g., a coin cell battery). The battery **426** may be configured to provide power to the control circuit **410**, the wireless communication circuit **414**, and/or to other low voltage circuitry of the faceplate remote control device **400**.

[0075] FIG. 6 depicts another example load control system that is configured as a lighting control system **500**. The lighting control system **500** may include various components that are associated with each other, and that are configured to communicate with one another, for instance via wireless communication. As shown, the lighting control system **500** includes a controllable light source **510** and a faceplate remote control device **520** that may be configured to control the controllable light source **510**. The controllable light source **510** and the faceplate remote control device **520** may include respective wireless communication circuits. For example, the controllable light source **510** may include a radio frequency (RF) transmitter, and the faceplate remote control device **520** may include an RF transceiver. The controllable light source **510** may be associated with the faceplate remote control device **520** during a configuration procedure of the lighting control system **500**, such that the controllable light source **510** may be configured to respond to one or more messages transmitted by the faceplate remote control device **520**.

[0076] The controllable light source **510** may include an integral lighting load (not shown) and an integral load regulation circuit (not shown). The integral lighting load of the controllable light source **510** may comprise, for example, an incandescent lamp, a halogen lamp, a compact fluorescent lamp, a light-emitting diode (LED) light engine, or other suitable light source. The integral load regulation circuit of the controllable light source **510** may comprise, for example, a dimmer circuit, a ballast circuit, or an LED driver circuit, for controlling the intensity of the

integral lighting load between a low-end intensity (e.g., approximately 1%) and a high-end intensity (e.g., approximately 100%). The faceplate remote control device **520** may be associated with the controllable light source **510**, such that the faceplate remote control device **520** may be operated to cause the integral load regulation circuit to control the integral lighting load. The controllable light source **510** may be installed in a ceiling-mounted downlight fixture **512**.

[0077] As shown, the downlight fixture **512** may be coupled to an AC power source **502** through a standard wall-mounted mechanical switch **504** (e.g., a toggle switch or a standard light switch). The mechanical switch **504** may include a toggle actuator **506**. The mechanical switch **504** may be opened and closed in response to actuations of (e.g., operation of) the toggle actuator **506**.

Accordingly, the controllable light source **510** may be turned on and/or off in response to actuations of the toggle actuator **506** of the mechanical switch **504**.

[0078] The controllable light source **510** may further include a control circuit (e.g., microprocessor) and a wireless communication circuit (e.g., comprising an RF receiver) that may be housed inside an enclosure portion of the controllable light source **510**. The control circuit may be configured to control the integral lighting load in response to one or more messages that are received at the wireless communication circuit (e.g., via RF signals **508**), such as messages received from the faceplate remote control device **520**. The controllable light source **510** may be configured similarly to the controllable light source **110** shown in FIG. 2, for example comprising a similar mechanical assembly and/or including similar electrical circuitry as.

[0079] The faceplate remote control device **520** may be configured to be attached to (e.g., mounted to) the mechanical switch **504**. For example, the faceplate remote control device **520** may be attached to the mechanical switch **504** in place of a standard faceplate or wall plate. In this regard, the faceplate remote control device **520** may replace a standard faceplate or wall plate that was previously attached to the mechanical switch **504**. The faceplate remote control device **520** may define an opening **522** through which the toggle actuator **506** of the mechanical switch **504** may protrude. As shown, the opening **522** is configured to permit the toggle actuator **506** to protrude through the opening **522** such that the toggle actuator **506** is operable, for example by a user of the lighting control system **500**, and in particular a user of the faceplate remote control device **520**.

[0080] As shown, the faceplate remote control device **520** may include one or more buttons, such as buttons **524**, **526**. The one or more buttons **524**, **526** may be configured to cause the wireless communication circuit of the faceplate remote control device **520** to transmit one or more messages. The one or more messages may be, for example, command messages that are transmitted to a device that is associated with the faceplate remote control device **520**, such as the controllable light source **510**. One or more of the buttons **524**, **526** may be operated, for example, to cause the controllable light source **510** to adjust the intensity of the integral lighting load, for instance when the mechanical switch **504** is closed. To illustrate, the faceplate remote control device **520** may be configured to transmit one or more messages (e.g., command messages), for example via RF signals **508**, to the controllable light source **510** in response to operation of one or more of the buttons **524**, **526**.

[0081] The faceplate remote control device **520** may be configured to operate as a state change device. For example, the faceplate remote control device **520** may be configured to transmit one or more messages, for example via RF signals **508**, in response to actuations of the toggle actuator **506** of the mechanical switch **504**. The one or more messages may be indicative of a change of state within the lighting control system **500**. For example, one or more messages may be indicative of a change of state of the toggle actuator **506** of the mechanical switch **504**. Such messages may be referred to as change of state messages, or as change of state signals, and may be interpreted by one or more devices that are associated with the faceplate remote control device **520**, such as the controllable light source **510**, as indications (e.g., commands) to turn on, turn off, dim, etc. respective lighting loads.

[0082] The lighting control system **500** may further include another load control device. For

example, as shown, the lighting control system **500** further includes a plug-in load control device **530**. The plug-in load control device **530** is plugged into an electrical outlet **536** that has two unswitched receptacles that are in electrical communication with the AC power source **502**. The lighting control system **500** further includes a floor lamp **532**. A standard light bulb **534** is installed in the floor lamp **532**. The floor lamp **532** is plugged into the plug-in load control device **530**, such that the plug-in load control device **530** may be operated to adjust the intensity of the light bulb **534** between a low end intensity (e.g., approximately 1%) and a high-end intensity (e.g., approximately 100%).

[0083] The plug-in load control device **530** may be associated with, and may be controlled by, the faceplate remote control device **520**. For example, the plug-in load control device **530** may cause the light bulb **534** to turn on or off in response to the receipt of one or more messages transmitted by the faceplate remote control device **520** (e.g., via RF signals **508**). This may allow the intensity of the light bulb **534** to be synchronized with that of the controllable light source **510**, for example. The plug-in load control device **530** may be configured to adjust the intensity of the light bulb **534**, for instance in response to one or more messages transmitted by the faceplate remote control device **520** in response to operation of one or more of the buttons **524**, **526**. The plug-in load control device **530** may include one or more buttons (not shown) that are configured to provide local control of the plug-in load control device **530**, for example to allow adjustment of the intensity of the light bulb **534**. Alternatively, the plug-in load control device **530** may comprise a tabletop load control device or a wall-mounted dimmer switch.

[0084] The lighting control system **500** may further include one or more other devices that are configured to transmit messages (e.g., via RF signals **508**) that may cause one or more load control devices of the lighting control system **500** to adjust corresponding lighting loads. For example, as shown, the lighting control system **500** further includes a battery-powered handheld remote control device **540**, an occupancy sensor **550**, and a daylight sensor **560** that may be configured to operate similarly to the battery-powered handheld remote control device **140**, the occupancy sensor **150**, and the daylight sensor **160**, respectively, of the lighting control system **100** shown in FIG. 2. The controllable light source **510** may be configured to control the intensity of the integral lighting load in response to one or more messages received from the handheld remote control device **540**, the occupancy sensor **550**, and/or the daylight sensor **560**, when the mechanical switch **504** is closed. The plug-in load control device **530** may be configured to control the intensity of the light bulb **534** in response to one or more messages received from the handheld remote control device **540**, the occupancy sensor **550**, and/or the daylight sensor **560**, while the plug-in load control device **530** is plugged into the electrical outlet **536**.

[0085] It should be appreciated that while the lighting control systems **100**, **500** illustrated in FIGS. 2 and 6, respectively, are described herein with reference to single-pole AC systems, that the apparatuses, features, and/or techniques described herein may be implemented in a three-way lighting system having two single-pole double-throw (SPDT) mechanical switches (e.g., standard three-way switches), to control a single electrical load. For example, a lighting control system in accordance with such a configuration may include two faceplate remote control devices, with one faceplate remote control device mounted to each SPDT switch. Moreover, the apparatuses, features, and/or techniques described herein may be implemented in a four-way lighting system, or in a lighting system having more control locations. Additionally, the apparatuses, features, and/or techniques described herein may be applied to direct-current (DC) distribution systems.

[0086] It should further be appreciated that the apparatuses, features, and/or techniques described herein are not limited to implementation in a faceplate remote control device that is configured to be mounted to a mechanical switch that is mounted in a wallbox. For example, one or more components of the faceplate remote control device may be integrated into a mechanical switch. Such a configuration may comprise a remote control switch device that may be deployed as a replacement for an existing wall-mounted mechanical switch, for instance. For example, the remote

control switch device may include an integral wireless communication circuit, and may be associated with one or more devices of a load control system, such as a load control device of a lighting control system. The remote control switch device may be configured to operate similarly to the faceplate remote control devices described herein. For example, the remote control switch device may be configured to transmit one or more messages (e.g., commands and/or change of state signals) to one or more devices that are associated with the remote control switch device, in response to operation of the toggle actuator of the remote control switch device. Such a remote control device may be configured, for example, with a standard toggle actuator, or with a Decora® or designer style toggle actuator.

Claims

1. A faceplate remote control device couplable to an electrical wallbox, the faceplate remote control device comprising: a front plate having a first surface and a second surface; a printed circuit board comprising a control circuit and a wireless communication circuit, wherein the printed circuit board is coupled to the first surface of the front plate; a user interface accessible from the second surface of the front plate; and a sensing device electronically connected to the control circuit of the printed circuit board, wherein the control circuit is configured to: receive a first input from the sensing device; and cause the wireless communication circuit to transmit a first message for controlling the electrical lighting load based on the first input from the sensing device.
2. The faceplate remote control device of claim 1, wherein the first input from the sensing device indicates a light intensity level, wherein the first message for controlling the electrical lighting load comprises an indication to adjust an amount of power being delivered to the electrical lighting load based on the indicated light intensity level.
3. The faceplate remote control device of claim 1, wherein the first input from the sensing device indicates an occupancy or vacancy condition, wherein the first message for controlling the electrical lighting load comprises an indication to turn on or off the electrical lighting load based on the occupancy or vacancy condition.
4. The faceplate remote control device of claim 1, wherein the control circuit is configured to: receive a second input from the user interface; and transmit a second message for controlling the electrical lighting load based on one or both of the first input from the sensing device and the second input from the user interface.
5. The faceplate remote control device of claim 1, wherein the faceplate remote control device is configured to be mountable to a wall-mounted light switch that is attached to a standard electrical wallbox.
6. The faceplate remote control device of claim 5, wherein the front plate comprises an opening, wherein the opening is configured such that, while the faceplate remote control device is mounted to the wall-mounted light switch, a toggle actuator of the wall-mounted light switch protrudes through the opening so that the toggle actuator can be actuated between an on position and an off position.
7. The faceplate remote control device of claim 6, wherein the printed circuit board comprises a toggle actuator detector circuit, wherein the control circuit is configured to: receive an indication of a position of the toggle actuator from the toggle actuator detector circuit, and cause the wireless communication circuit to transmit a second message for controlling the electrical lighting load based on the received indication of the position of the toggle actuator.
8. The faceplate remote control device of claim 7, wherein the toggle actuator detector circuit comprises a mechanical tactile switch, or an infrared (IR) transmitter and IR receiver.
9. The faceplate remote control device of claim 7, wherein the toggle actuator detector circuit comprises a sliding member and a conductive pad, wherein the sliding member comprises an electrically conductive wiper, wherein the sliding member configured to move concurrently with

the toggle actuator, so that the sliding member is in contact with the conductive pad while the toggle actuator is in the on position, but the sliding member is not in contact with the conductive pad while the toggle actuator is in the off position.

- 10.** The faceplate remote control device of claim 1, wherein the user interface comprises one or more control buttons.
 - 11.** The faceplate remote control device of claim 1, wherein the user interface comprises a visual display configured to display an indication of an amount of power being delivered to the electrical lighting load.
 - 12.** The faceplate remote control device of claim 1, wherein the sensing device comprises at least one of an occupancy sensing circuit, a vacancy sensing circuit, or a daylight sensing circuit.
 - 13.** The faceplate remote control device of claim 1, wherein the front plate comprises a lens, wherein the sensing device comprises a passive infrared detector that is operable to receive infrared energy through the lens.
 - 14.** The faceplate remote control device of claim 1, further comprising a battery electronically connected to the control circuit, wherein the battery is configured to provide power to the control circuit, the sensing device, and the wireless communication circuit.
 - 15.** The faceplate remote control device of claim 1, wherein the first message for controlling the electrical lighting load comprises an indication of a state change associated with the electrical lighting load.
 - 16.** The faceplate remote control device of claim 1, wherein the first message for controlling the electrical lighting load comprises an indication to increase or decrease an amount of power being delivered to the electrical lighting load.
 - 17.** The faceplate remote control device of claim 1, wherein the first message for controlling the electrical lighting load comprises an indication to turn the electrical lighting load on or off.
 - 18.** The faceplate remote control device of claim 1, wherein the front plate comprises a plurality of attachment members, wherein the attachment members are configured to enable the faceplate remote control device to be coupled with the electrical wallbox using an adapter plate.
 - 19.** The faceplate remote control device of claim 1, further comprising a programming button, wherein the programming button is disposed on the second surface of the front plate, wherein the control circuit is configured to initiate a process to associate the faceplate remote control device with a lighting control system upon actuation of the programming button.
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